



**ROBOTICS**  
**ENGR – UH 4330**  
**Fall, 2025**

Experiment: **Lab 8 – Trapezoidal Profile**

Due date: **16 / NOV / 2025**

Section Number: 001

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## 1. Introduction

In lab 3, a PID controller was used to command the joint position of a robotic actuator. When a step input is applied directly to the position controller, the joint attempts to move instantaneously toward the final position. This introduces high acceleration, high torque demand, vibrations, and large transient currents on the actuator.

To address these issues, this lab introduces a trajectory generator, a module placed before the controller to ensure that reference commands change smoothly. Instead of stepping directly to a new position, the trajectory generator produces a smooth time-based path that respects limits on acceleration and velocity. This greatly improves system stability and reduces mechanical stress.

One of the most commonly used functions for generating smooth joint references is the trapezoidal velocity profile. It divides the motion into three phases: constant acceleration, constant velocity, and constant deceleration. This type of trajectory guarantees limited jerk, bounded acceleration, and physically achievable actuator motion.

The goal of this lab is to implement a trapezoidal velocity profile generator in MATLAB, verify its correctness, and analyse the resulting position, velocity, and acceleration curves.

## 2. Theory

A trapezoidal velocity profile moves a joint from an initial position  $q_i$  to a final position  $q_f$  over a prescribed time interval  $t_f$ . The velocity profile consists of three segments:

1. Acceleration phase
2. Constant-velocity (cruise) phase
3. Deceleration phase

The corresponding position function  $q(t)$  is defined piecewise as:

$$q(t) = \begin{cases} q_i + \frac{1}{2}\ddot{q}_c t^2 & 0 \leq t \leq t_c \\ q_i + \dot{q}_c t_c (t - t_c/2) & t_c < t \leq t_f - t_c \\ q_f - \frac{1}{2}\ddot{q}_c (t_f - t)^2 & t_f - t_c < t \leq t_f. \end{cases}$$

The corner time  $t_c$  is computed from:

$$t_c = \frac{q_i - q_f + \dot{q}_c t_f}{\dot{q}_c}$$

The constant acceleration required is:

$$\ddot{q}_c = \frac{\dot{q}_c^2}{q_i - q_f + \dot{q}_c t_f}$$

To ensure a physically valid trapezoidal profile, the selected cruise velocity must satisfy:

$$\frac{|q_f - q_i|}{t_f} < |\dot{q}_c| \leq \frac{2|q_f - q_i|}{t_f}$$

If this inequality is not satisfied, the motion becomes triangular or impossible.

### 3. Methodology

The lab required implementing a MATLAB function that:

- Accepts initial and final joint positions
- Accepts a chosen cruise velocity and final trajectory time
- Computes one point  $q(t_k)$  per function call
- Uses the manual's equations exactly
- Ensures the trapezoidal conditions are met
- Returns an error if the trajectory cannot be generated

A separate test script was used to:

- Sample the trajectory at N equally spaced points
- Compute numerical velocity and acceleration
- Plot position, velocity, and acceleration curves

All simulations were performed using MATLAB.

#### 4. MATLAB Implementation and Results from Test Cases

The MATLAB code implementation can be found in the documents attached to this report submission.

Test Cases were generated for  $N = 500$  trajectory points for various values of cruise velocity in the range of 20-50 while keeping the initial and final velocities at 10 and 90 m/s, respectively, time at 4 seconds.

To satisfy these conditions a trajectory plot would only be feasible if the cruise velocity falls within the following range:

$$22.5 < \dot{q}_c \leq 45$$

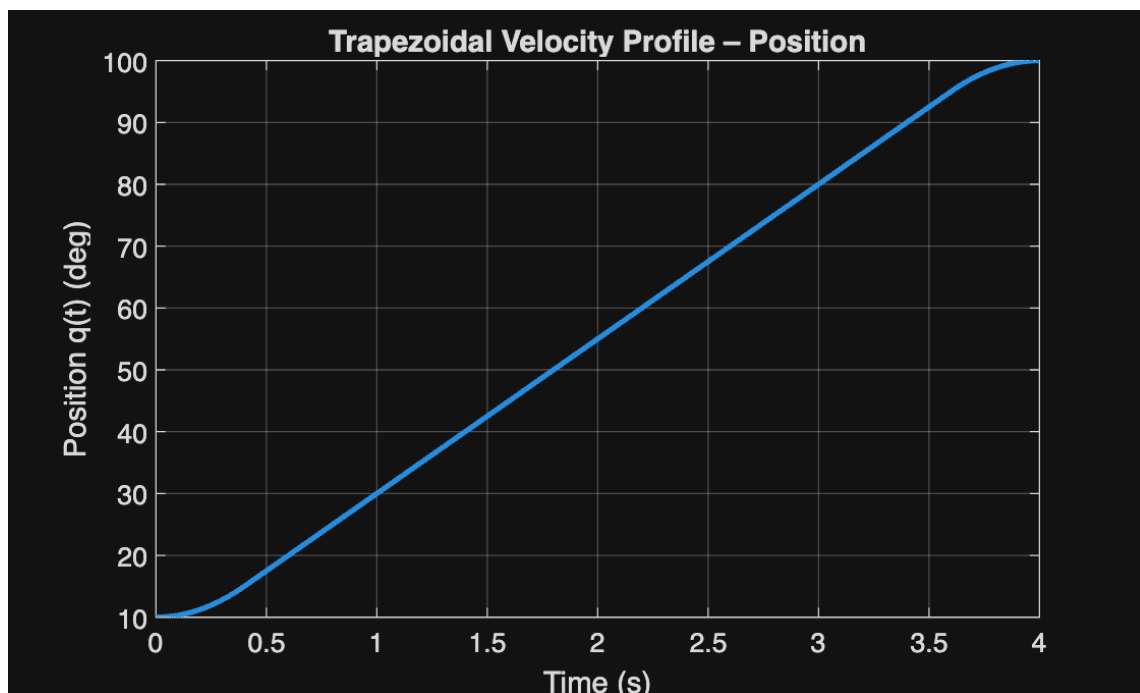
##### Test Case 1: $\dot{q}_c = 20$ rev/min

Since  $\dot{q}_c = 20$  rev/min falls below the minimum required cruise velocity, a trajectory cannot be created.

Cannot generate trajectory: cruise velocity does not satisfy condition.

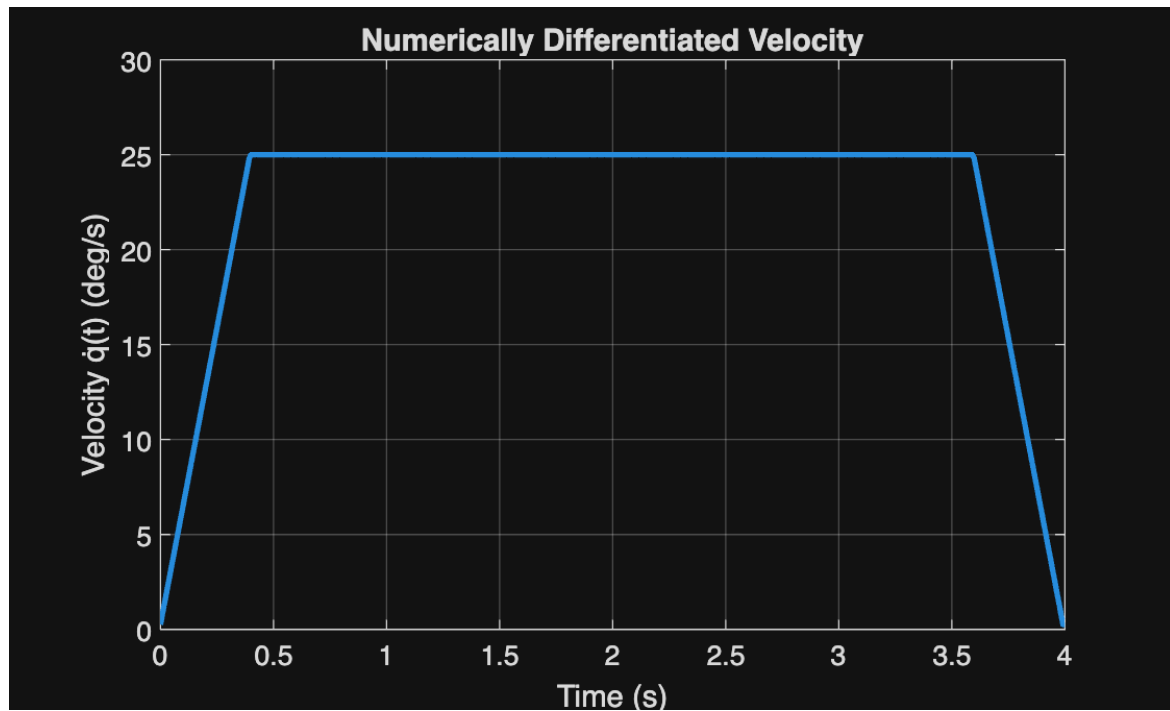
##### Test Case 2: $\dot{q}_c = 25$ rev/min

Figure 2.1 – Position Profile



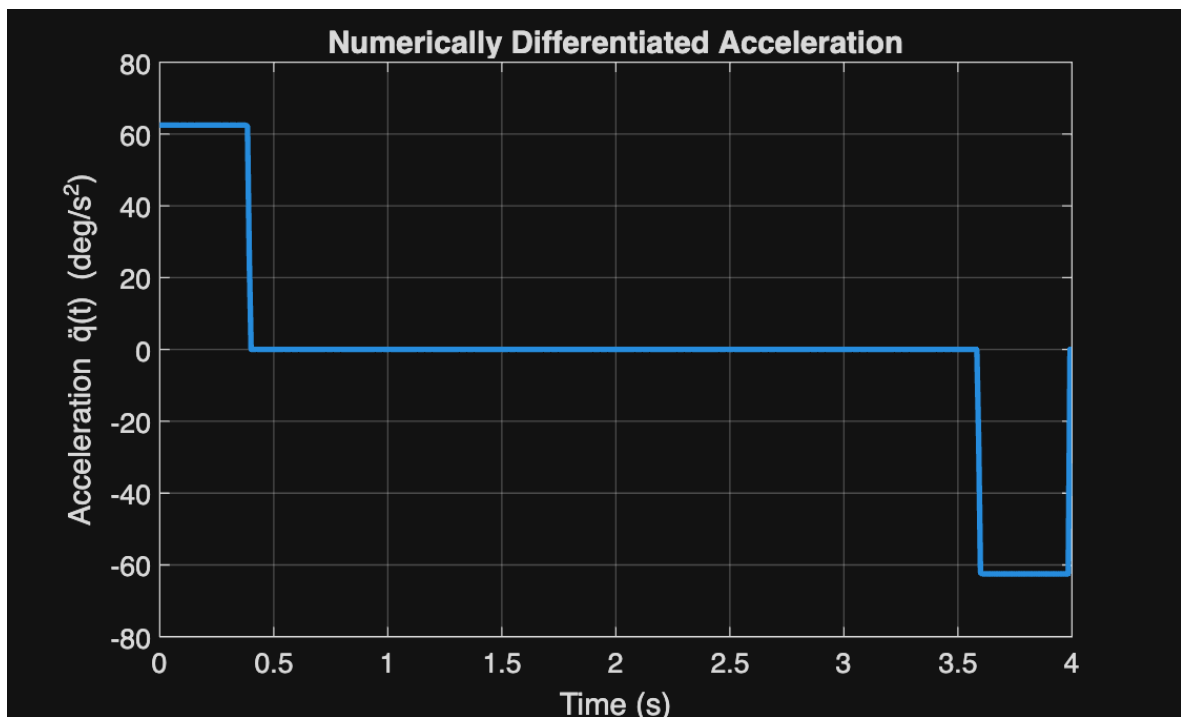
The position curve shows a smooth S-shape, confirming continuous velocity and limited acceleration.

**Figure 2.2 – Velocity Profile**



A clear trapezoidal shape is observed, with acceleration, constant-velocity, and deceleration segments.

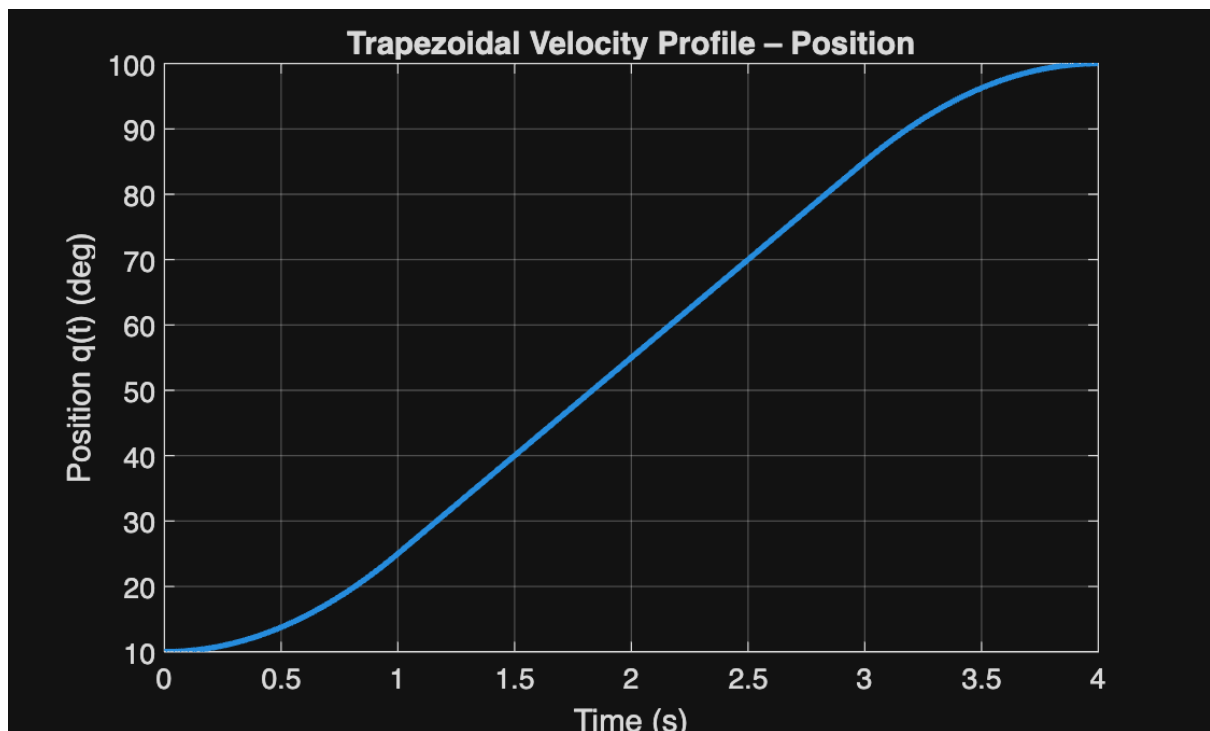
**Figure 2.3 – Acceleration Profile**



Acceleration is piecewise constant, with  $+\ddot{q} = 60$  and  $-\ddot{q} = -60$  phases.

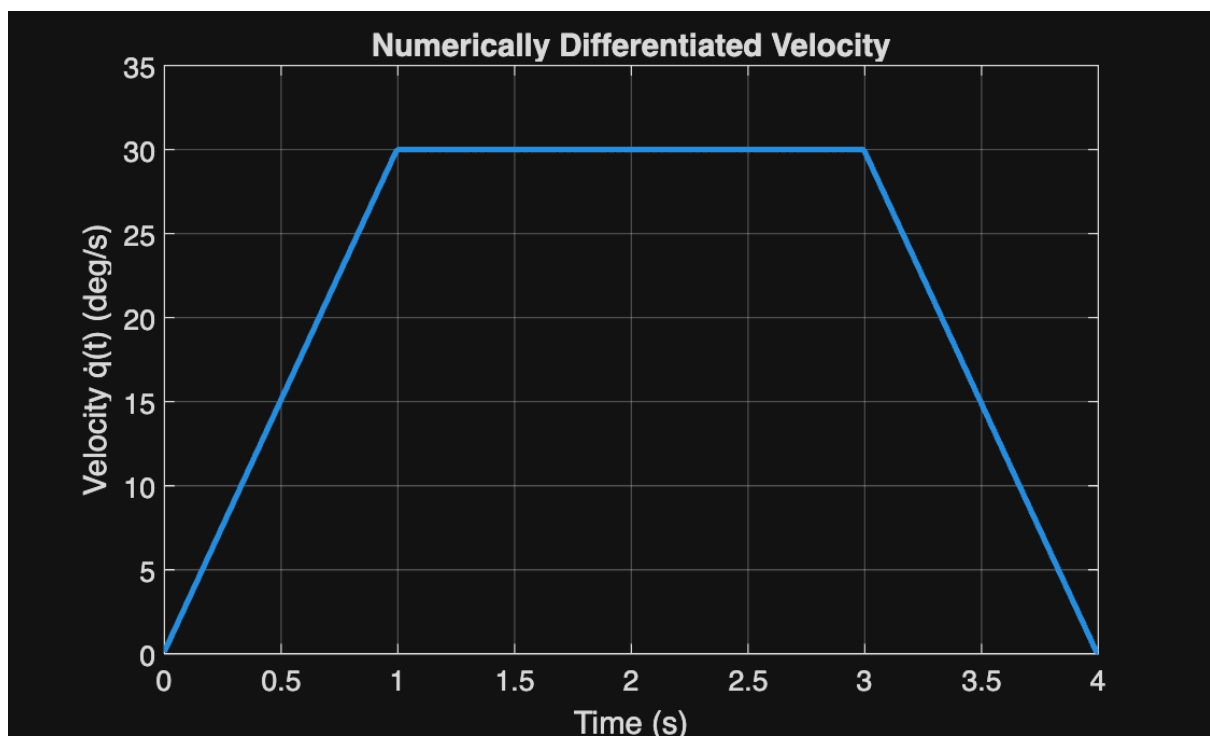
**Test Case 3:  $\dot{q}_c = 30$  rev/min**

**Figure 3.1 – Position Profile**



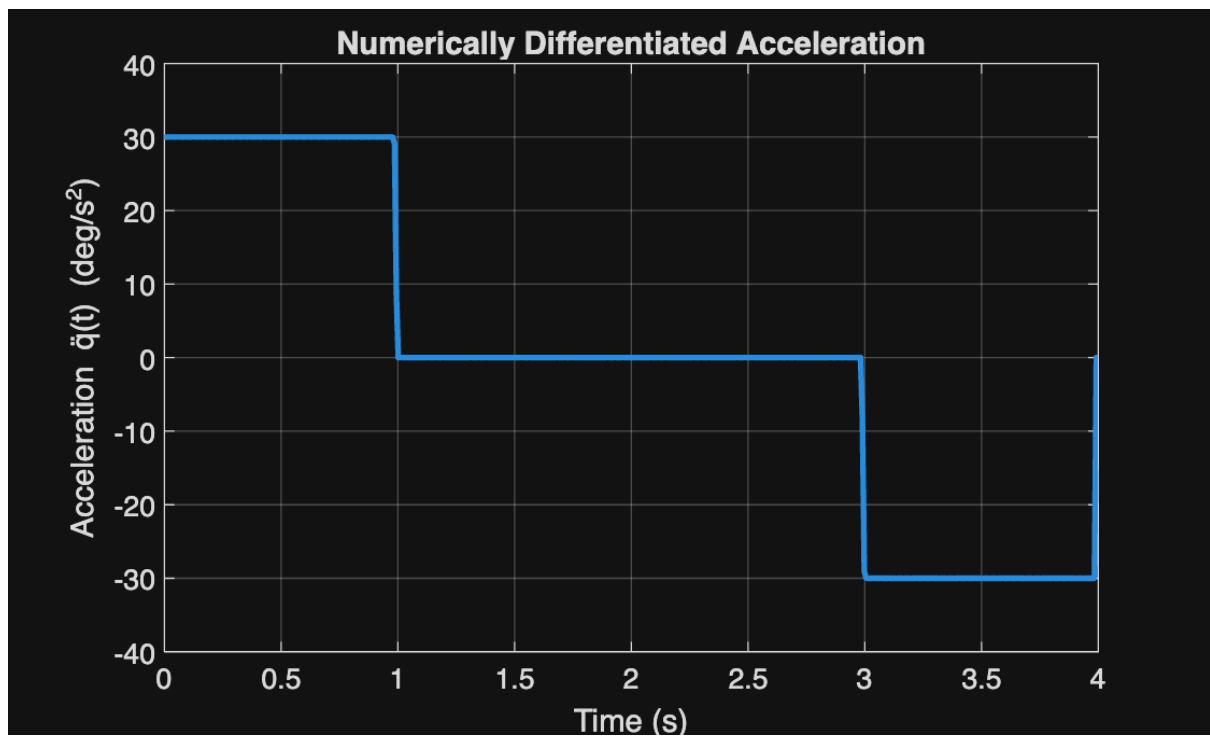
The position curve shows a smooth S-shape, confirming continuous velocity and limited acceleration.

**Figure 3.2 – Velocity Profile**



A clear trapezoidal shape is observed, with acceleration, constant-velocity, and deceleration segments.

**Figure 3.3 – Acceleration Profile**

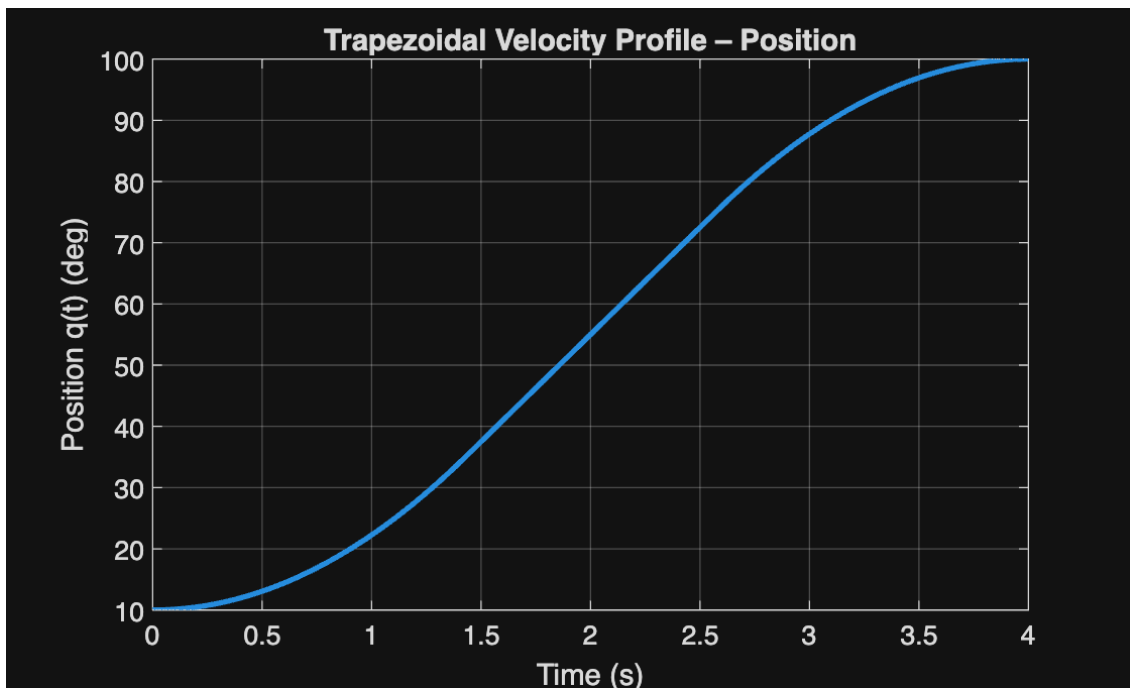


Acceleration is piecewise constant, with  $+\ddot{q} = 30$  and  $-\ddot{q} = -30$  phases.



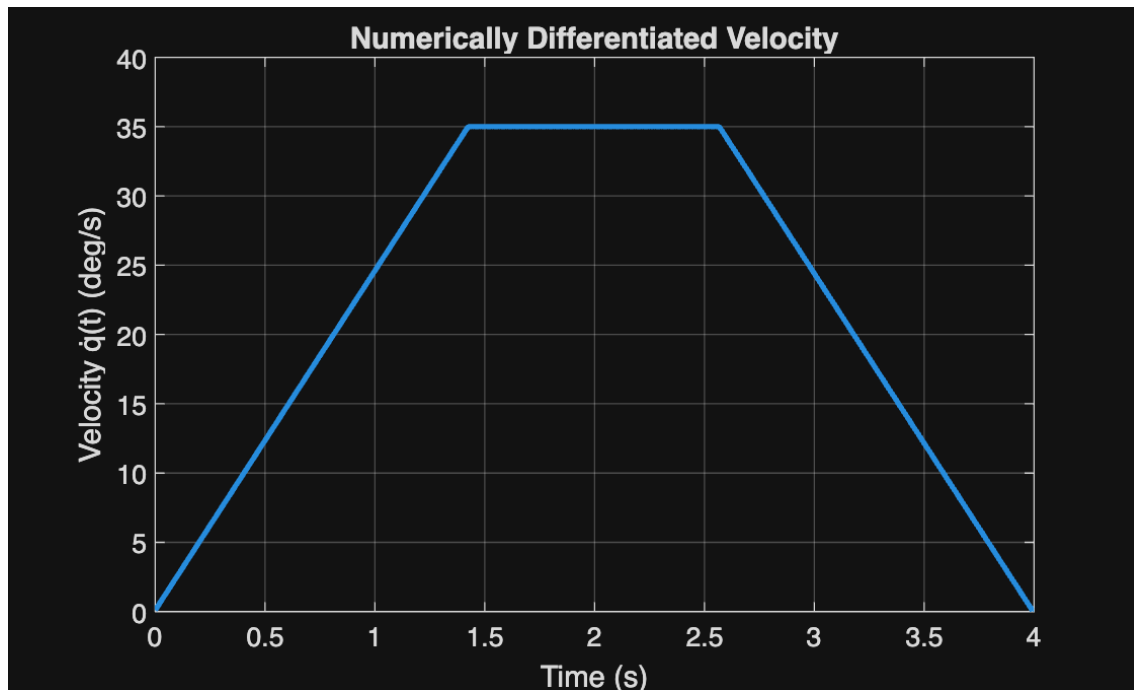
Test Case 4:  $\dot{q}_c = 35$  rev/min

Figure 4.1 – Position Profile



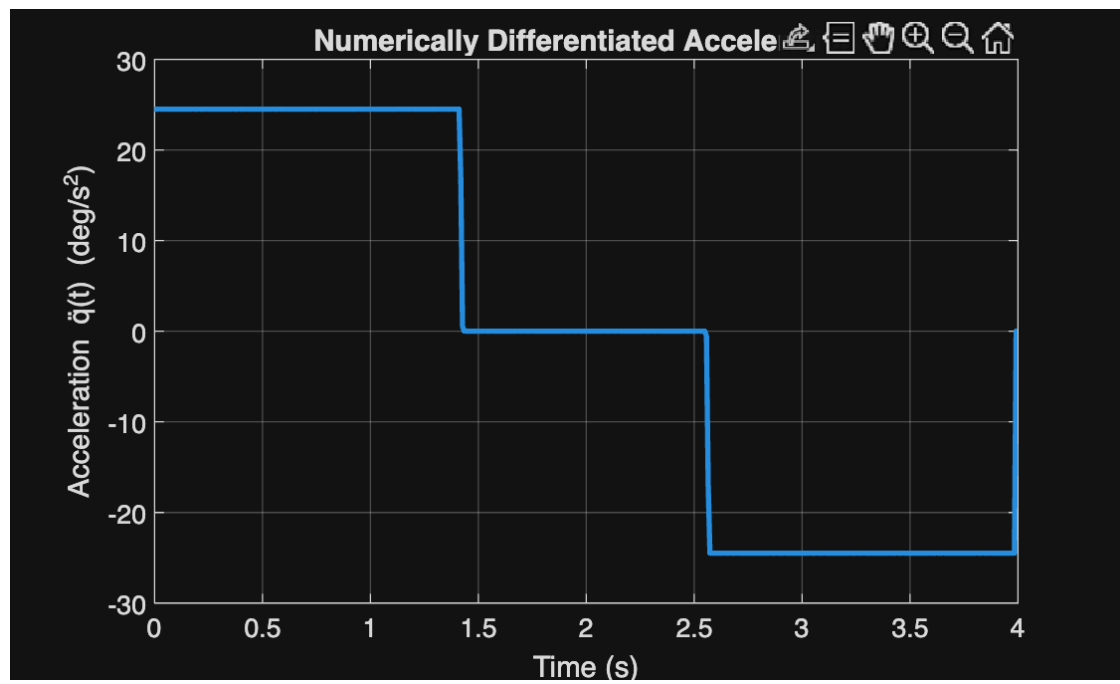
The position curve shows a smooth S-shape, confirming continuous velocity and limited acceleration.

Figure 4.2 – Velocity Profile



A clear trapezoidal shape is observed, with acceleration, constant-velocity, and deceleration segments.

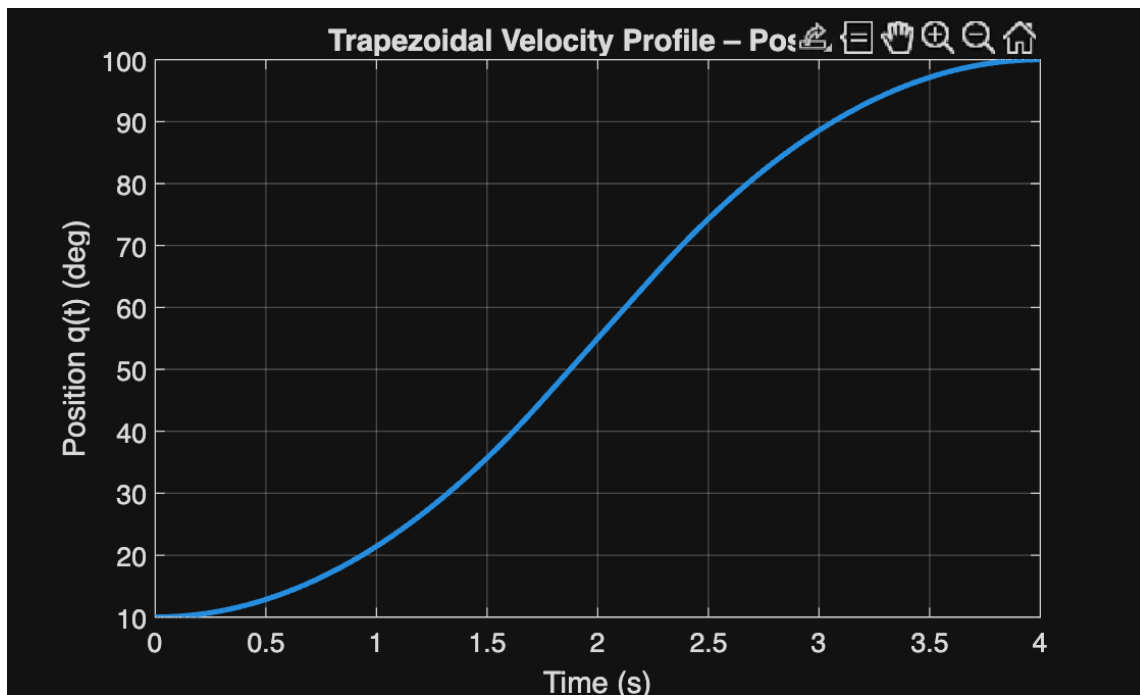
**Figure 1.3 – Acceleration Profile**



Acceleration is piecewise constant, with  $+\ddot{q} = 24.5$  and  $-\ddot{q} = -24.5$  phases.

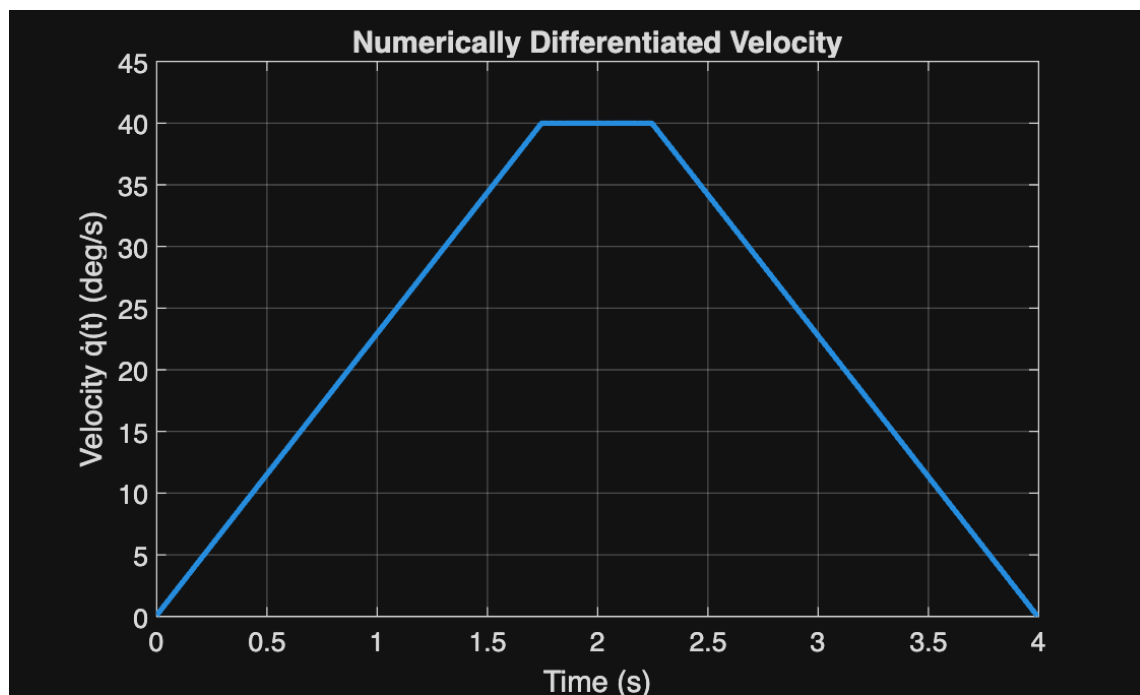
Test Case 5:  $\dot{q}_c = 40$  rev/min

Figure 5.1 – Position Profile



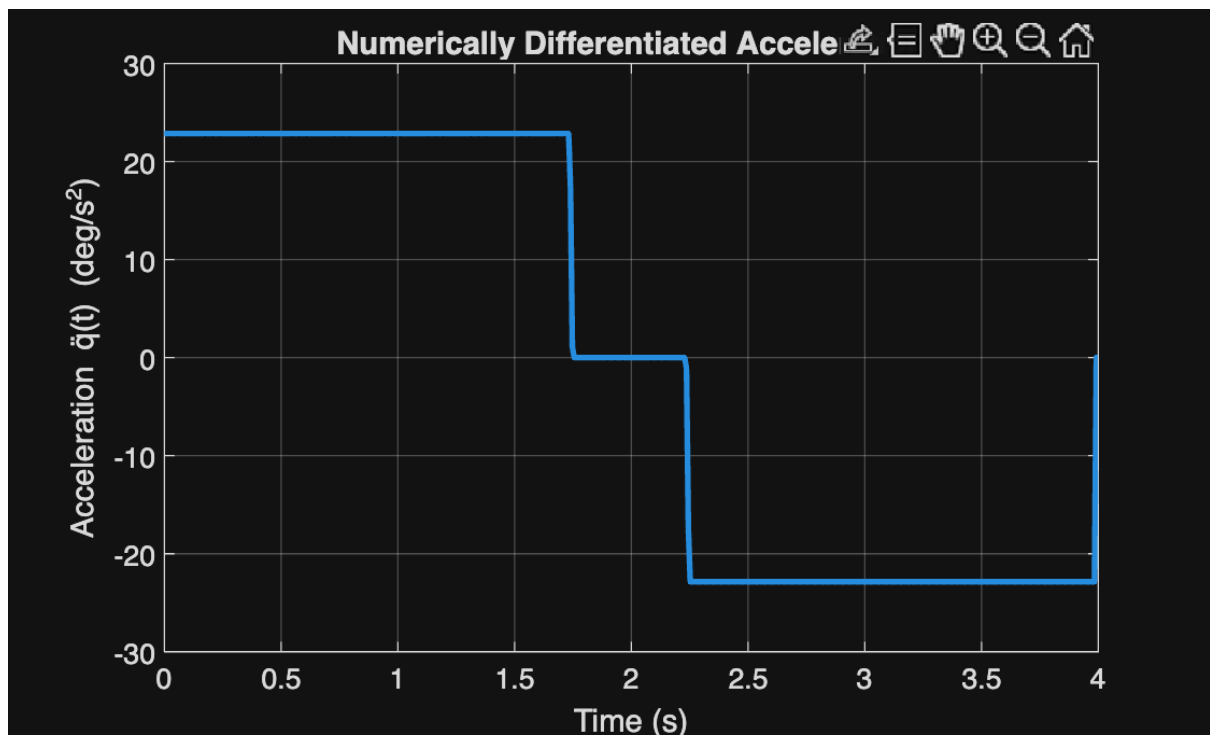
The position curve shows a smooth S-shape, confirming continuous velocity and limited acceleration.

Figure 5.2 – Velocity Profile



A clear trapezoidal shape is observed, with acceleration, constant-velocity, and deceleration segments.

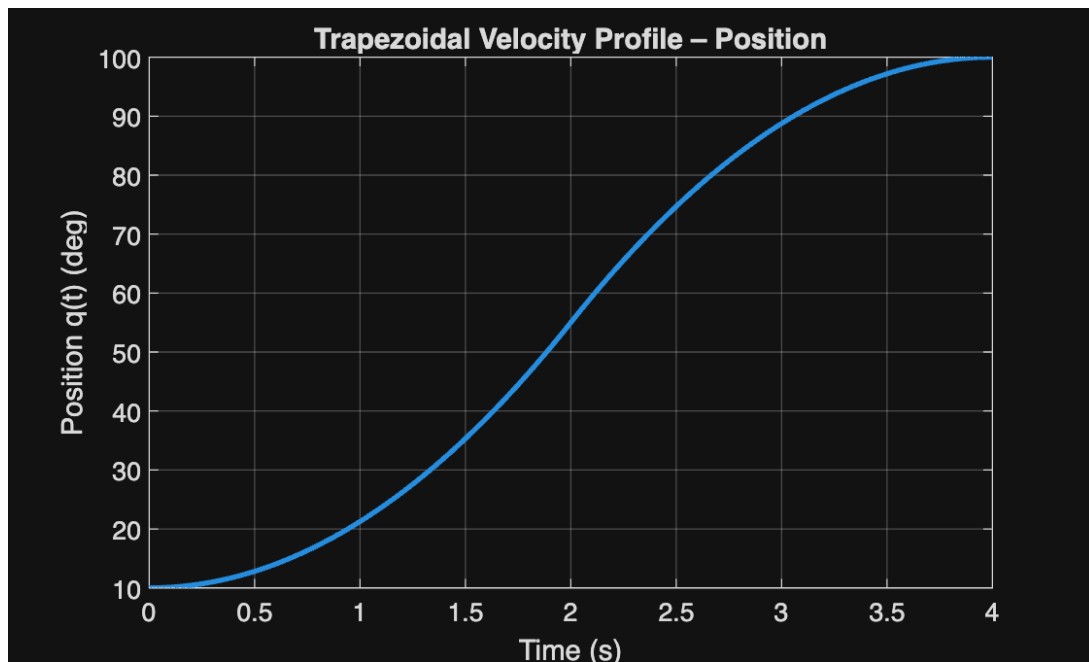
**Figure 5.3 – Acceleration Profile**



Acceleration is piecewise constant, with  $+\ddot{q} = 22.875$  and  $-\ddot{q} = -22.875$  phases.

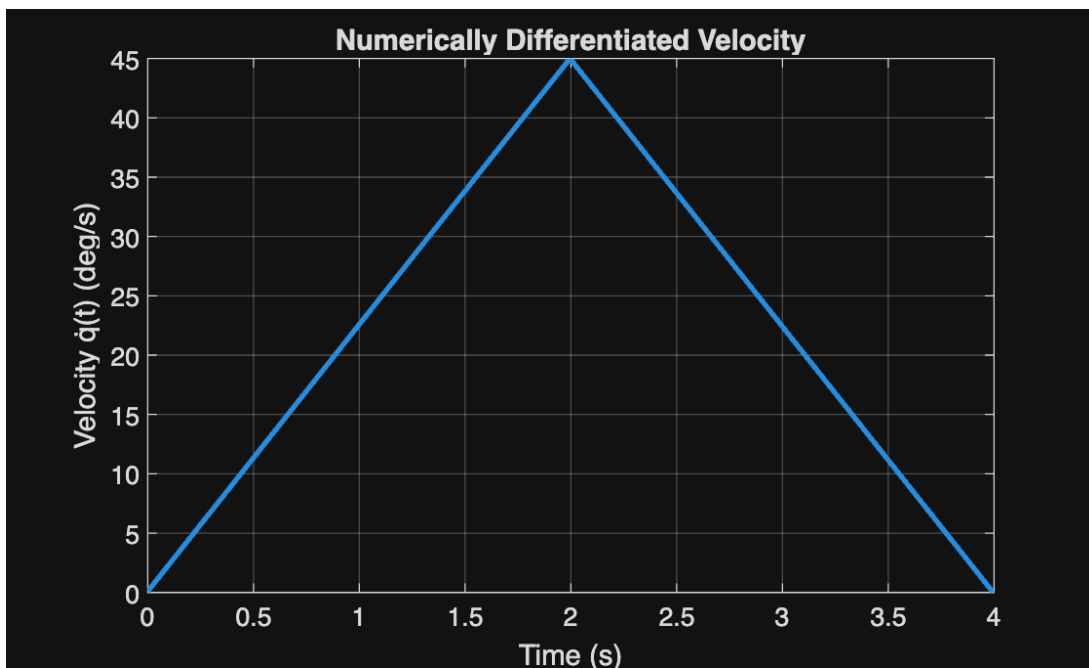
Test Case 6:  $\dot{q}_c = 45$  rev/min (maximum cruise velocity that generates a trajectory)

Figure 6.1 – Position Profile



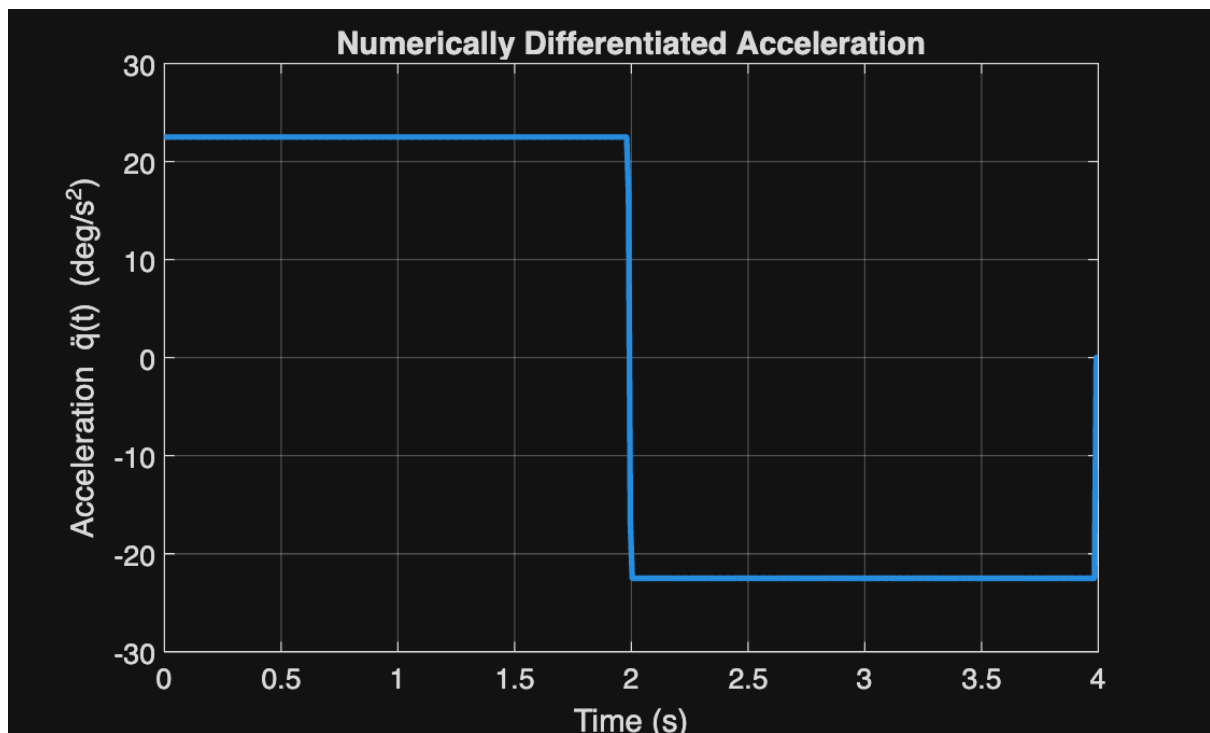
The position curve shows a smooth S-shape, confirming continuous velocity and limited acceleration.

Figure 6.2 – Velocity Profile



A clear triangular shape is observed, with acceleration and deceleration segments, but no constant-velocity segment. This is because the cruise velocity is at the upper limit of the possible value.

**Figure 6.3 – Acceleration Profile**



Acceleration is piecewise constant, with  $+\ddot{q} = 22.5$  and  $-\ddot{q} = -22.5$  phases. Note the acceleration changes suddenly from positive to negative and is never zero because there is no constant velocity at maximum cruise velocity.

**Test Case 7:  $\dot{q}_c = 50$  rev/min**

Since  $\dot{q}_c = 50$  rev/min is beyond the maximum required cruise velocity, a trajectory cannot be created.

Cannot generate trajectory: cruise velocity does not satisfy condition.

## 5. Discussion

The trapezoidal velocity profile successfully generates smooth motion between the initial and final joint positions. By enforcing bounded acceleration, the trajectory removes discontinuities and high-frequency behaviour that would otherwise excite vibrations in the mechanical system.

The results confirm that:

- The acceleration and deceleration phases are symmetric.
- A constant-velocity region exists only when the cruise velocity satisfies the manual's inequality and is less than the maximum value. At the maximum allowed cruise velocity, there is no constant velocity and consequently the acceleration changes sign abruptly.
- Selecting a cruise velocity close to the lower limit results in a longer constant velocity / cruise time and shorter acceleration and deceleration periods.
- Selecting a cruise velocity at the upper boundary eliminates the cruise region, producing a triangular velocity profile.

The numerical differentiation produces clean velocity and acceleration curves, validating the correctness of the implemented trajectory generator.

## 7. Conclusion

A trapezoidal velocity profile trajectory generator was implemented successfully using the equations provided in the lab manual. The generated trajectory ensures smooth motion with limited acceleration and eliminates issues associated with step commands, such as vibrations and high torque demand. The MATLAB results confirm the theoretical expectations, demonstrating the validity and usefulness of trapezoidal velocity profiles for robotic joint control.